

ASCII MASTER FLOW – PANEL 1/12: Why England first: conjunction

CAPITAL -> finances fixed investment before returns

COAL + IRON -> energy and machinery base

LABOUR + FOOD -> enclosure and agricultural improvement

MARKETS + LAW -> scale, contracts and commercial security

MUST REMEMBER: Britain's first industrial breakthrough joined coal and iron, agricultural and demographic change, credit, transport, imperial markets and institutions; invention alone does not explain timing.

ASCII MASTER FLOW – PANEL 2/12: Cotton innovation chain

1764 -> Hargreaves: spinning jenny

1769 -> Arkwright water frame; Watt improvement era

1785 -> Cartwright: power loom

1793 -> Whitney: cotton gin in the United States

ASCII MASTER FLOW – PANEL 3/12: Putting-out to factory

DOMESTIC SYSTEM -> dispersed tools and household rhythm

FACTORY -> concentrated machinery and supervision

STEAM -> production freed from muscle and water limits

TIME DISCIPLINE -> wage labour organised by the clock

ASCII MASTER FLOW – PANEL 4/12: Transport multiplier

CANALS + ROADS -> cheaper movement of bulk goods

STEAM BOATS -> faster water transport

PASSENGER RAILWAY 1830 -> national networks deepen

RESULT -> coal, iron, engineering and markets reinforce each other

ASCII MASTER FLOW – PANEL 5/12: Industrial class society

CAPITALISTS -> own productive machinery and seek profit

WAGE WORKERS -> lack tools and depend on employers

CITY -> housing, sanitation and health become public issues

POLITICS -> class grievance enters organisation and franchise

ASCII MASTER FLOW – PANEL 6/12: Gender and childhood ledger

WOMEN -> central factory workers and household earners

CHILDREN -> cheap labour under weak protection

CONDITIONS -> long hours, low wages and insecurity

CAUTION -> experience varied by trade, place and household

ASCII MASTER FLOW – PANEL 7/12: Social response ladder

1802 -> first Factory Act begins regulation

1824 -> anti-union laws repealed

1830s-40s -> Chartism demands political rights

LONG RESULT -> laissez-faire yields to labour evidence

ASCII MASTER FLOW – PANEL 8/12: Uneven diffusion map

ENGLAND -> earliest machine-and-factory surge

FRANCE/BELGIUM -> continental growth after 1815

GERMANY/USA -> later powerful acceleration

RUSSIA/JAPAN -> compressed, state-assisted routes

ASCII MASTER FLOW – PANEL 9/12: Britain versus Japan

BRITAIN -> private profit joins capital, coal and markets

JAPAN 1868 -> state treats industry as national security

BRITAIN -> domestically generated invention sequence

JAPAN -> railways, roads, cotton and silk promoted from above

ASCII MASTER FLOW – PANEL 10/12: Railways change with context

INDUSTRIALISER -> integrates markets and heavy industry

CONTINENTAL CATCH-UP -> links state policy and national market

SETTLER FRONTIER -> opens land and displaces indigenous peoples

COLONY -> moves exports and administrative power toward ports

CLOSE DISTINCTION: Textiles, steam, factory discipline and railways were connected mechanisms but had different chronologies and effects, while industrialisation outside Britain followed multiple state, labour and resource paths.

ASCII MASTER FLOW – PANEL 11/12: Industry-to-empire pressure

OUTPUT RISES -> raw-material demand expands

SCALE RISES -> overseas markets become more valuable

CAPITAL ACCUMULATES -> investment outlets are sought

RIVALRY INTENSIFIES -> empire gains economic weight

EVIDENCE LIMIT: Output and connectivity expanded alongside urban crowding, class conflict, gendered labour, ecological damage and colonial deindustrialisation; progress and harm require spatially differentiated evidence.

ASCII MASTER FLOW – PANEL 12/12: Mixed-blessing answer spine

GAIN -> productivity, transport and cheaper goods

COST -> labour insecurity, slums, women and child exploitation

RESPONSE -> unions, Factory Acts and Chartism

VERDICT -> broad delayed gains, concentrated immediate costs